ML Summer Bridge: Week 3 Study Notes

Machine Learning Class

Summer 2026

Week 3: Can a Line Help Us Predict?

Last week you made two plots: one showing height as a function of father's height, and one showing height as a function of mother's height. You probably noticed that both plots suggested some kind of relationship - taller parents tended to have taller children. This week, we are going to think a little more carefully about *what kind* of relationship this is, and more importantly, whether we can use it to make predictions.

Before we go further, let's improve our plot a little bit. Last week we used kind='line', which connects the dots in the order they appear in our table. That's not quite what we want here, since the order of our rows doesn't mean anything special - person 3 isn't "after" person 2 in any meaningful sense. Instead, we want to see the points scattered in space so we can look for a pattern. We can do this with kind='scatter':

```
data.plot(x='Father\'s Height', y='Height', kind='scatter')
plt.show()
```

Go ahead and make this plot, along with the analogous one for mother's height. Take a good look at the dots.

Here is a question for you to think about, and I really do want you to think about it rather than skip ahead: if you had to describe the pattern of dots in one sentence, what would you say? Is it a curve? A cloud with no shape at all? Or does it look like the dots are hovering roughly around some straight line?

Now suppose a new family moves to town. We meet the father, and he happens to be 71 inches tall. We haven't met his child yet - maybe the child hasn't been born, maybe they're just not home when we visit. But we would still like to guess how tall this child will turn out to be (or already is). Here is the key question of this entire week:

Using only the information in our table (and nothing else) could we make a reasonable guess for this new child's height?

Take a moment before reading on. Look back at your scatter plot. Find 71 on the horizontal axis (father's height). Even though no single point in our table has a father's height of exactly 71, you probably have a rough sense of "around where" the corresponding height should land, just by looking at the nearby dots.

That instinct - "the point should land somewhere near where the other nearby points are" - is exactly the intuition behind using a *line* to make predictions. If we could draw a single straight line that passes reasonably close to all of our dots, then for any father's height we are given (even one we've never seen before, like 71), we could simply find where that height would sit on our line and use the corresponding vertical value as our predicted height.

Try this concretely: take a ruler, a piece of paper, or even just your finger on the screen, and draw a straight line through your scatter plot that seems to best capture the trend of the dots. It doesn't have to be perfect, and it doesn't have to pass exactly through any of the points. Once you've drawn it, find where a father's height of 71 would land on your line, and read off your predicted height. Write down your guess.

Here are a few things worth noticing about what you just did:
1. None of our original data points needs to actually sit exactly on the line. In fact, it's very likely that *none* of them do.
2. Different people in this class, drawing their own "by eye" line, will probably come up with slightly different lines - and slightly different predictions for our height-71 father. That should feel a little unsatisfying. Is there a way to make everyone agree on the "correct" line, rather than everyone eyeballing their own?
3. We used only father's height here, but we also have mother's height and gender available. Could a line still help us if we wanted to use more than one of these features at once?

We are not going to answer any of these three questions this week - we will save the details for when we are all together in the classroom. But I do want you to sit with them for a bit, because they are exactly the questions that motivate the topic we will spend the first real chunk of the course on: *linear regression.* In short, linear regression is a systematic (and non-arguable!) way of finding "the best" line through a set of points, so that everyone - no matter who draws it - ends up with the exact same line and the exact same predictions.

One more thing to ponder before next week: what do you think "best" should even mean when we're talking about a line through a scatter of points? If two different lines are both "pretty close" to the data, how would you decide which one is actually better? There's no need to come up with a precise answer - a sentence or two of your own intuition is more than enough. Bring your thoughts to our first class, and we'll build the real definition together.

Great work this week. This one was more about thinking than about typing code, and that's on purpose - the coding will come soon enough!